

Enterprise AI Development Challenge Casebook

Digital Innovation | SDC MSc Innovation Management

Powered by Kimi AI

May 2026

Introduction

The case companies and all numbers are made up for educational purposes. Any resemblance to real enterprises is purely coincidental.

This casebook is designed for the Digital Innovation course (SDC MSc Innovation Management), featuring two enterprises and two AI development challenges.

Each case provides rich business data, system context, and challenge descriptions. Student teams have 4 weeks to build runnable prototypes using the Kimi AI model.

Cases are calibrated for MSc Innovation Management level, balancing technical innovation with business analysis. Suitable for predominantly non-technical teams (5 students, 1-2 with technical background).

Case 1: Smart Warehouse Sorting & Dispatch System

Company Background

"A Large Pharmaceutical Logistics Enterprise" (hereinafter "the Enterprise") is one of China's leading pharmaceutical distribution companies, with over 40 years in the industry and annual revenue exceeding RMB 100 billion.

The Enterprise operates a nationwide "integrated wholesale-retail-storage-distribution" pharmaceutical logistics service system, with 128 modern logistics centers, 856 terminal distribution stations, 2.65 million square meters of GSP-standard warehouse space, annual throughput exceeding 90 million cases, covering 28 provinces and over 94% of administrative regions.

In 2025, the Enterprise achieved revenue of RMB 138.76 billion, serving 12,800 public hospitals, 5,832 chain pharmacies, 196,500 independent pharmacies, and 357,400 primary healthcare institutions.

Business Data

Metric	Data
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Metric	Data
Logistics Centers	128 (incl. 6 central warehouses)
Distribution Stations	856
Warehouse Area	2.65m sqm (GSP-standard)
Annual Throughput	>90m cases
Product Lines (SKU)	423,600 (Western/TCM 58,700, devices 231,400, TCM 79,500)
Partner Manufacturers	8,600
Public Hospital Clients	12,800
Retail Pharmacy Clients	>250,000 (incl. 5,832 chain)
B2B Registered Users	>580,000
Next-Day Delivery Rate	>88%

Challenge Description

The Enterprise's warehousing system faces these core challenges:

- **SKU complexity:** Among 423,600 product lines, ~38% are near-expiry managed items with varying temperature requirements (ambient, cool, cold, frozen), making sorting rules highly complex.
- **Order fragmentation:** The B2B platform sees >36,000 daily active customers, mostly small-batch multi-SKU orders averaging 3-5 items per order, yet total daily orders exceed 90,000.
- **Time pressure:** The >88% next-day delivery promise requires same-day sorting and dispatch; peak periods (e.g., flu season) can spike order volume by 280%.
- **Labor bottleneck:** Sorting relies heavily on manual labor; peak periods require 2.5x temporary staffing, with ~0.35% error rate and annual return losses exceeding RMB 9 million.
- **Inventory turnover:** Some pharmaceutical products have near-expiry scrap rates of ~4.8%, urgently requiring smart dispatch systems to optimize FIFO strategy.

Task Requirements

Design and develop a smart warehouse sorting & dispatch system with these core features:

- **Smart wave allocation:** Auto-generate sorting waves based on order characteristics, product attributes, and warehouse layout to optimize picking paths.

- Dynamic inventory scheduling: Real-time monitoring of bin-level inventory with intelligent replenishment/reallocation recommendations.
- Near-expiry alert: Generate expiry warnings and promotion suggestions based on product shelf life and outbound frequency.
- Visual dashboard: Display real-time sorting progress, labor load, and exception alerts across all logistics centers.
- Simulation: Provide "what-if" analysis to simulate system performance under different order volumes.

Deliverables

- Runnable system prototype (web app or demo)
- System architecture documentation
- One concrete business scenario demo (e.g., sorting dispatch simulation for a regional warehouse during flu season peak)

Case 2: Pharmaceutical Demand Forecasting & Intelligent Replenishment

Company Background

Same enterprise as Case 1. The Enterprise operates China's largest out-of-hospital (OOH) sales network, with OOH sales accounting for >68%. Its B2B e-commerce platform achieved 2025 sales revenue of RMB 19.847 billion (+19.86% YoY).

The platform has >580,000 registered users, 340,000 active users, and 95.1% terminal customer satisfaction. It integrates intelligent search, AI customer service, smart procurement, and automated order verification.

Business Data

Metric	Data
B2B Annual Sales	RMB 19.847b (+19.86%)

Metric	Data
Registered Users	>580,000 (clinics, chain & independent pharmacies)
Active Users	340,000
Customer Satisfaction	95.1%
Daily Active Customers	>36,000
Self-Operated SKUs	380,000
Partner Manufacturers	8,600
Franchise Pharmacies	>28,600
Volume Procurement Rights	3,247 (286 exclusive)

Challenge Description

The Enterprise's pharmaceutical supply chain management faces these challenges:

- High demand volatility: Pharmaceutical demand is affected by seasonal diseases (flu, allergy seasons), public health emergencies, and policy changes (volume procurement), resulting in low forecast accuracy.
- Long-tail SKUs: Among 380,000 SKUs, 78% are low-frequency items; some sell fewer than 100 boxes annually, yet stockouts lead to customer churn.
- Stockouts and overstock coexist: Hot-selling products frequently stock out (annual loss ~RMB 420m), while slow-moving items pile up with turnover exceeding 165 days for some categories.
- Decentralized replenishment: 28,600 franchise pharmacies each make independent ordering decisions, lacking unified smart restock recommendations; most rely on experience-based ordering.
- Policy shock: National volume procurement policies have reduced selected drug prices by ~51% on average, causing volatile sales that break traditional forecasting models.

Task Requirements

Design and develop a demand forecasting & intelligent replenishment system:

- Multi-factor demand forecasting: Integrate seasonality, epidemiological data, policy changes, and historical sales to forecast 14-30 day demand by category.

- Intelligent replenishment recommendations: Generate personalized restock lists for hospitals, chain pharmacies, and independent stores with suggested order quantities and optimal timing.
- Inventory health diagnosis: Analyze terminal inventory structure, identify slow-moving, fast-moving, and near-expiry items, and provide reallocation suggestions.
- Policy impact simulation: Model how volume procurement and other policy changes affect specific drug demand.
- Alerts & notifications: Proactively push alerts for impending stockouts or inventory overages.

Deliverables

- Runnable forecasting & replenishment system prototype
- One SKU-level forecast demo (e.g., 3-month sales trend prediction vs. actual for a chronic disease medication)
- Business feasibility analysis (target users, value proposition, go-to-market strategy)

How to Use

Case Selection

Each team selects one of the two cases. Choose based on your team's technical background and interest.

Development Tips

- Leverage Kimi's code generation, data analysis, and NLP capabilities fully.
- Don't aim for completeness — focus on polishing 1-2 core features with excellent UX.
- Treat AI as a "collaborator" not a "black box" — show how you used AI tools to accelerate development.
- Focus on business feasibility: your solution needs technical viability AND a clear answer to "who will pay for this?"

Timeline

Date	Milestone
May 19	Course launch, team formation & case selection
May 26	Proof-of-concept demo
Jun 3	Mid-term review & feature demo
Jun 10	Final Demo Day presentation
Jun 12	Written submission
Jun 17-18	Oral examination

Good Luck!

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